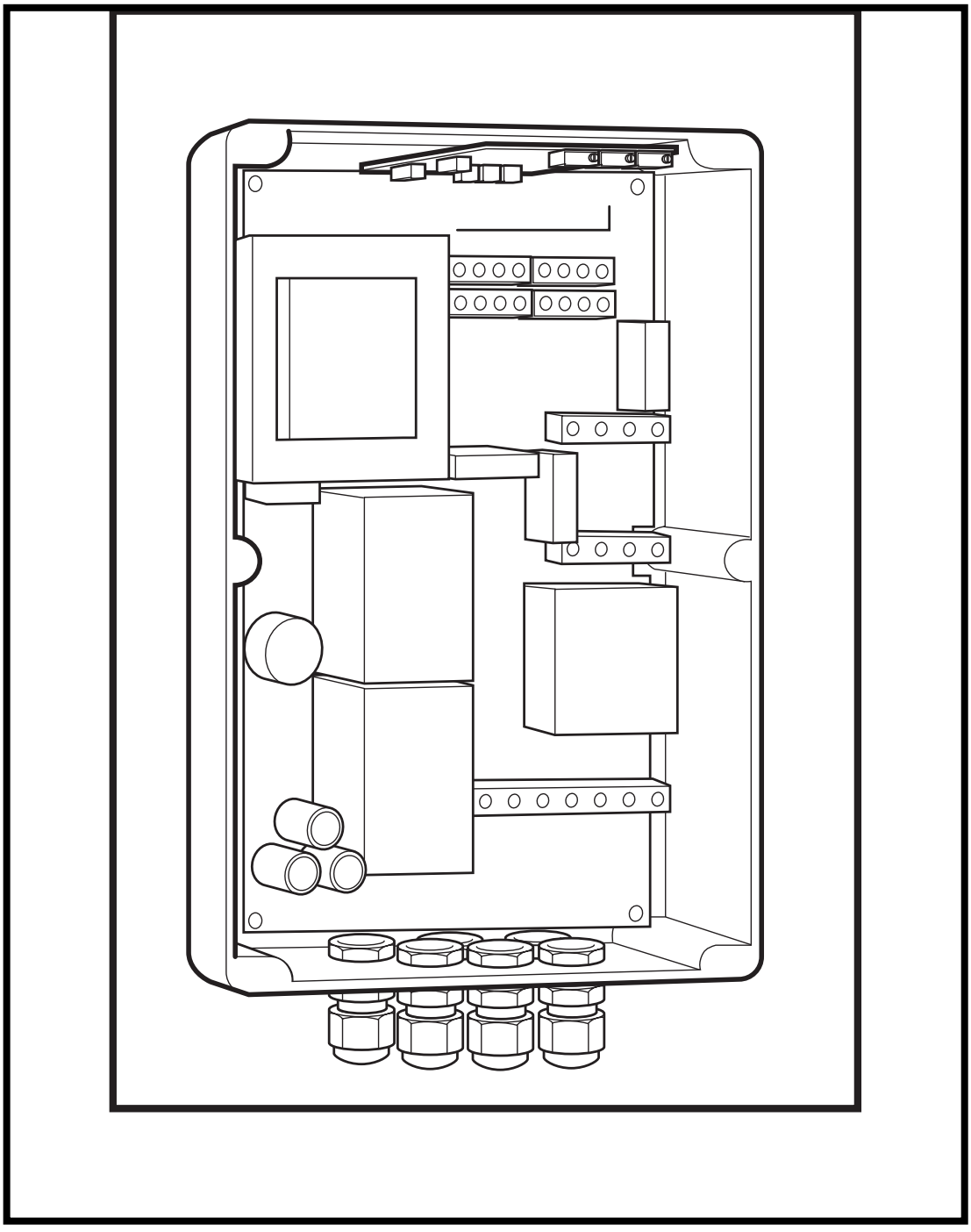
ALBANY
DOOR SYSTEMS
NOMAFA SCHIEFFER JANSEN

ACS 30
CONTROL SYSTEM



User manual

English version

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Manual No: 6410V0016 Issue: 11/2002

ALBANY DOOR SYSTEMS
NOMAFA SCHIEFFER JANSEN

Part. Nr. 98-9999-95
Issue 2002-01-01

FÖRSÄKRAN OM ÖVERENSSTÄMMELSE

Vi Albany Door Systems AB
(leverantörens namn)

Box 509, S-301 80 Halmstad, Sweden
(adress)

försäkrar under eget ansvar att produkten

Rapid Roll Type 220,320,330,350
420,450,520,550,660,910
Rapid Tech Type 750, 770
Rapid LiftType 940
(namn, typ eller modell, parti- eller serienummer, eventuellt ursprung och antal enheter)

Som omfattas av denna försäkran är
i överensstämmelse med följande standarder
eller andra regelgivande dokument
EN 292-1 1992.01.29, EN 292-2 1992.01.29,
EN 954-1 1994.11, EN 50 081-2 1992.01,
EN 50 082-2 1992.01, EN 60 204-1:1998.10.20
(titel och/eller beteckning och utgivningsdatum för standarden (standarderna)eller andra regelgivande dokument)

(om tillämpligt)
enligt villkoren i direktiv
98/37/EEC, 73/23/EEC, 93/68/EEC, 92/31/EEC
89/336/EEC

Halmstad 2002.01.01
(plats och datum för utförandet)

Håkan Larsson
Product development manager

DECLARATION OF CONFORMITY

We Albany Door Systems AB
(suppliers name)

Box 509, S-301 80 Halmstad, Sweden
(address)

declare under our sole responsibility that the product

Rapid Roll Type 220,320,330,350
420,450,520,550,660,910
Rapid Tech Type 750, 770
Rapid LiftType 940
(name, type or model, lot, batch or serial number, possibly sources and numbers of items)

to which this declaration relates is in conformity with
the following standard(s) or other normative
document(s)
EN 292-1 1992.01.29, EN 292-2 1992.01.29,
EN 954-1 1994.11, EN 50 081-2 1992.01,
EN 50 082-2 1992.01, EN 60 204-1:1998.10.20
(title and/or number and date of issue of the standard(s) or other normative document(s))

(if applicable)
following the provisions of Directive
98/37/EEC, 73/23/EEC, 93/68/EEC, 92/31/EEC
89/336/EEC

Halmstad 2002.01.01
(Place and date of issue)

Håkan Larsson
Product development manager

DÉCLARATION DE CONFORMITÉ

Nous Albany Door Systems AB
(nom du fournisseur)

Box 509, S-301 80 Halmstad, Sweden
(adresse)

déclarons sous notre seule responsabilité que le produit

Rapid Roll Type 220,320,330,350
420,450,520,550,660,910
Rapid Tech Type 750, 770
Rapid LiftType 940
nombre
(nom, type ou modèle, no. de lot, d'échantillon ou de série, éventuellement sources et d'exemplaires)

auquel se réfère cette déclaration est conforme à la
(aux) norme(s) ou autre(s) document(s) normatif(s)
EN 292-1 1992.01.29, EN 292-2 1992.01.29,
EN 954-1 1994.11, EN 50 081-2 1992.01,
EN 50 082-2 1992.01, EN 60 204-1:1998.10.20
(titre et/ou no. et date de publication de la (des) norme(s) ou autre(s) document(s) normatif(s))

(le cas échéant)
conformément aux dispositions de Directive
98/37/EEC, 73/23/EEC, 93/68/EEC, 92/31/EEC
89/336/EEC

Halmstad 2002.01.01
(Lieu et date)

Håkan Larsson
Product development manager

KONFORMITÄTSERKLÄRUNG

Wir Albany Door Systems AB

(Name des Anbieters)

Box 509, S-301 80 Halmstad, Sweden

(Anschrift)

erklären in alleiniger Verantwortung, daß das Produkt

Rapid Roll Type 220, 320,330,350,
420,450,520,550,660,910
Rapid Tech Type 750,770
Rapid Lift Type 940

(Bezeichnung, Typ oder Modell, Los-, Chargen- oder Seriennummer, möglichst Herkunft und Stückzahl)

auf das sich diese Erklärung bezieht, mit der/den folgenden Norm(en) oder Dokument(en) übereinstimmt.

EN 292-1 1992.01.29, EN 292-2 1992.01.29,
EN 954-1 1994.11, EN 50 081-2 1992.01,
EN 50 082-2 1992.01, EN 60 204-1:1998.10.20

(Titel und/oder Nummer sowie Ausgabedatum der Norm(en) oder der anderen normativen Dokumente)

(falls zutreffend)

Gemäß den Bestimmungen der Richtlinie
98/37/EEC,73/23/EEC,93/68/EEC,92/31/EEC
89/336/EEC

Halmstad 2002.01.01

(Ort und Datum der Ausstellung)

Håkan Larsson
Product development manager

DECLARACIÓN DE CONFORMIDAD

Albany Door Systems AB

(nombre del suministrador)

Box 509, S-301 80 Halmstad, Sweden

(dirección)

declaramos bajo nuestra exclusiva responsabilidad la conformidad del producto

Rapid Roll Type 220,320,330,350
420,450,520,550,660,910
Rapid Tech Type 750, 770
Rapid LiftType 940

(nombre, tipo o modelo, nº de lote, de muestra o de serie, en su caso: proceedencia y número de ejemplares)

al que se refiere esta declaración, con la(s) norma(s) u otro(s) documento(s) normativo(s).

EN 292-1 1992.01.29, EN 292-2 1992.01.29,
EN 954-1 1994.11, EN 50 081-2 1992.01,
EN 50 082-2 1992.01, EN 60 204-1:1998.10.20

(titulo y/o número y fecha de la(s) norma(s) u otro(s) documento(s) normativo(s))

(en su caso)

de acuerdo con las disposiciones de Directiva
98/37/EEC, 73/23/EEC, 93/68/EEC, 92/31/EEC
89/336/EEC

Halmstad 2002.01.01

(lugar y fecha)

Håkan Larsson
Product development manager

OVEREENKOMSTIGHEIDSVERKLARING

Wij Albany Door Systems AB

(naam van leverancier)

Box 509, S-301 80 Halmstad, Sweden

(adres)

verklaren geheel onder eigen verantwoordelijkheid dat het produkt

Rapid Roll Type 220,320,330,350
420,450,520,550,660,910
Rapid Tech Type 750, 770
Rapid LiftType 940

(naam, type of model, charge-, partij- of serienummer, mogelijk herkomst en aantallen van de artikelen)

waarop deze verklaring betrekking heeft, in overeenstemming is met de volgende norm(en) of ander(e) normatie(f)(ve) documen(en)

EN 292-1 1992.01.29, EN 292-2 1992.01.29,
EN 954-1 1994.11, EN 50 081-2 1992.01,
EN 50 082-2 1992.01, EN 60 204-1:1998.10.20

(titel en/of nummer en uitgavedatum van de norm(en) of ander(e) normatie(f)(ve) document(en))

(indien van toepassing)

volgens de bepalingen van Richtlijn.
98/37/EEC, 73/23/EEC, 93/68/EEC, 92/31/EEC
89/336/EEC

Halmstad 2002.01.01

(plaats en datum van uitgifte)

Håkan Larsson
Product development manager

ACS 30 control system

Foreword and reading guide

This user manual gives you a presentation of the Albany Door Systems ACS 30 control system. You will learn the control systems functions, and how they are used.

Read and acquaint yourself with the user manual. This will make it easier for you to quickly locate the correct section when you work with the Albany Door Systems ACS 30 system.

The following is an overview of the manuals content and presentation:

1. Safety instructions: Relevant safety instructions, and the importance of heeding them.

2. Use: Use of the ACS 30 control system, and the adjustment of certain parameters.

3. A-pos: Contains instructions for adjustments of the A-pos and some trouble shoting.

4. Maintenance: Contains instructions for preventive maintenance, and maintenance instructions.

5. System overview: Standard capabilities of the ACS 30 control system, and available options.

Appendix: Address information and service record.

Note

The information in this manual may be outdated by alterations in later versions of machine and program line.

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Address information and service record	Appendix

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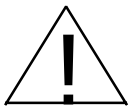
1. Safety instructions



WARNING

The instructions in this manual must be followed

Contact a Albany Door Systems distributor if you are uncertain of anything.



WARNING

General instructions

1. Check the delivery carefully to ensure that no components are damaged. If damage is found, contact your Albany Door Systems distributor.
2. If the product is modified without Albany Door System's explicit acknowledgement, manufacturer product liability is null and void.
3. Only Albany Door Systems-approved spare parts may be used.
4. Read user instructions carefully before the equipment is used.
5. Personnel must be properly trained before they may use the equipment.

NOTE !

The door fulfils the directives according to "Declaration of Conformity", which is enclosed in the delivery.

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CAUTION

Note these points carefully before installation

1. The product is designed for permanent installation.
2. The product must be installed according to installation instructions.
3. Installation must be carried out by qualified personnel.



DANGER

Electrical installation

1. The product must be joined to earth according to installation instructions.
2. High voltage- and signal cables must be kept separate.



WARNING

Follow these instructions in daily use

1. Keep the equipment clean.
2. Do not press buttons with sharp objects.
3. Albany Door Systems doors have a high built-in safety level. Regardless of this, you should not stand in or near the door opening more than necessary.
4. Do not run the door if the sideframe doors are open (if the door is equipped with openable sideframes).

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DANGER

Caution at repair and servicing

1. Qualified personnel must conduct repair and service.
2. Before working on the door, make sure that all scaffolds, ladders, etc. meet safety requirements and are in good condition.
3. Before maintenance work begins, power supply must be switched off at a lockable main switch, and the switch marked with a sign indicating that work is in progress. Then check the power supply with a voltmeter, to ensure that the electrical cabinet has no power.

NOTE !

Follow these instructions at disassembly and scrapping

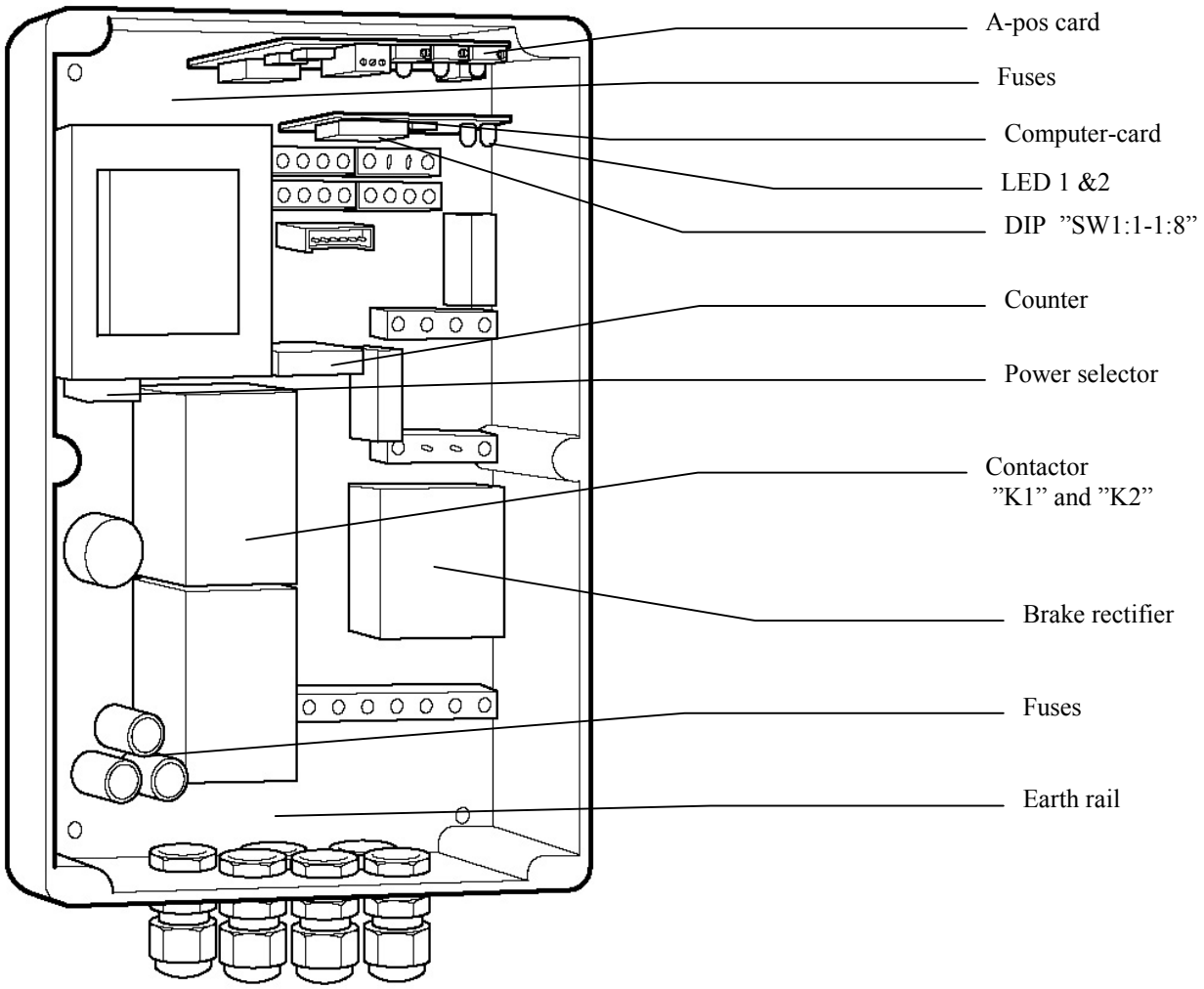
1. At disassembly, follow installation instructions in reverse order.
2. Recycling and/or disposal of the product and its components must be done in accordance with the prevailing local regulations.

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2. Use

Configuration

The following refers to the ACS 30 controlbox:

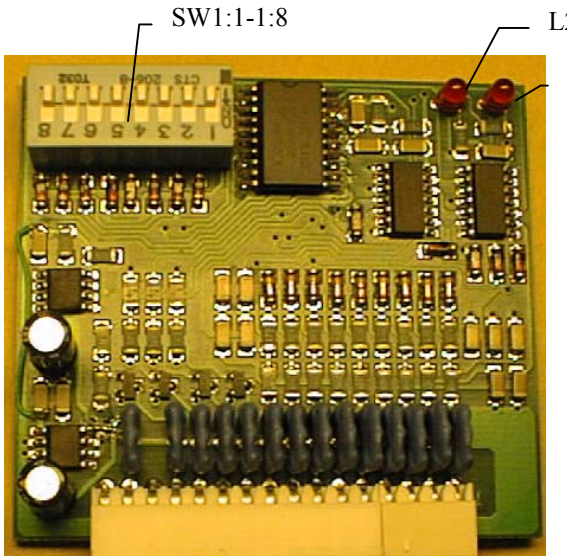


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Switch settings (SW1)

The switch (SW1) is located on the function-card, under the cover of the ACS 30

SW1:	ON	OFF
SW1:1	Pneumatic safety edge	Electric safety edge
SW1:2	Hold-To-Run Down mode (installation)	Normal mode
SW1:3	Automatic closing after safety edge hit.	Not automatic closing after safety edge hit.
SW1:4	Delayed opening (2 sec.)	Normal mode.



The time for automatic closing can be set from 1-60 seconds according to the table to the right.

	Terminal 3,4	Terminal 5,6
Any SW 1:5-1:8 <>0FF	↕	↷
All SW 1:5-1:8 = OFF	↑	↓

SW1:8	SW1:7	SW1:6	SW1:5	Time (s)
off	off	off	off	↑/↓
off	off	off	on	1
off	off	on	off	2
off	off	on	on	3
off	on	off	off	4
off	on	off	on	5
off	on	on	off	7
off	on	on	on	10
on	off	off	off	13
on	off	off	on	16
on	off	on	off	19
on	off	on	on	22
on	on	off	off	25
on	on	off	on	30
on	on	on	off	45
on	on	on	on	60

↕ = Open/close function.

↷ = Automatic closing 1 sec to 60 sec.

↑ = Open function.

↓ = Close function.

Users manual - ACS 30 control system
Manual No. 6410v0016 Issue: 11/2002
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LED functions

At the computer card there are 2 LED´s with the following function.

L2	L1	Description
Off	Flash	Flashes during start-up (3s)
Off	Flash	If an opening signal is on during start up.
Off	Off	Normal operation. If door still not moving emergency stop may be activated. Check the buttons and release if necessary.
On	On	Safety edge error (electric)
Off	On	Pneumatic safety edge test not OK
Off	On	Photocell test not OK
Off	On	Safety edge hit, and SW1:3 off
On	Off	Safety edge activated.
On	Off	Safety edge activated and door at top position. Then giving close signal L2 is flashing and L1 is on.
Flash	On	Break away activated. Hold-To-Run mode waiting for Emergency Stop to be pressed, and reset by closing. During reset-closing only L1 is on.
Flash	On	Photocells switched transmitter / receiver, can not be reset by Hold-To-Run mode, necessary to be connected in a right way, then push Emergency Stop and run the door.
Flash	Flash	Run time error, reset by switching power off / on.

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Photocell in doorline

When the photocell in doorline is disconnected by the A-POS card the receiver is lit up the whole time it is disconnected.
To make an proper check that the photocell is working, test it when the door is in open position.

Disengagement

When necessary (e.g. during a power failure) the door can be opened manually as follows:

A) Balanced doors

1. Release the clutch by turning the red handle.
2. Grip the bottom beam and raise the door to the necessary height.
3. Reset the clutch by turning back the red handle. Move the door by hand until it stops.

To close the door manually, release the clutch as above and lower the door to its closed position. Reset the clutch by turning back the red handle. Move the door by hand until it stops.

B) Unbalanced doors

Insert a hand crank at the back of the motor and crank the doorblade up or down to the desired position.

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Break-away

Check the following on the door after the door have been hit and stopped by the break-away system:

220 door

1. Check that the plastic bottom beam is not broken.
2. Check that the foam attached to the door is still fixed to the bottom beam.
3. Bring back the bottom beam and the door blade in the slot in the side frame.
4. Reset the door by pressing the emergency stop and release it again. Close the door with the pushbutton in hold-to-run mode (comes automatic).
5. Run the door.
6. Check the electrical safety edge.

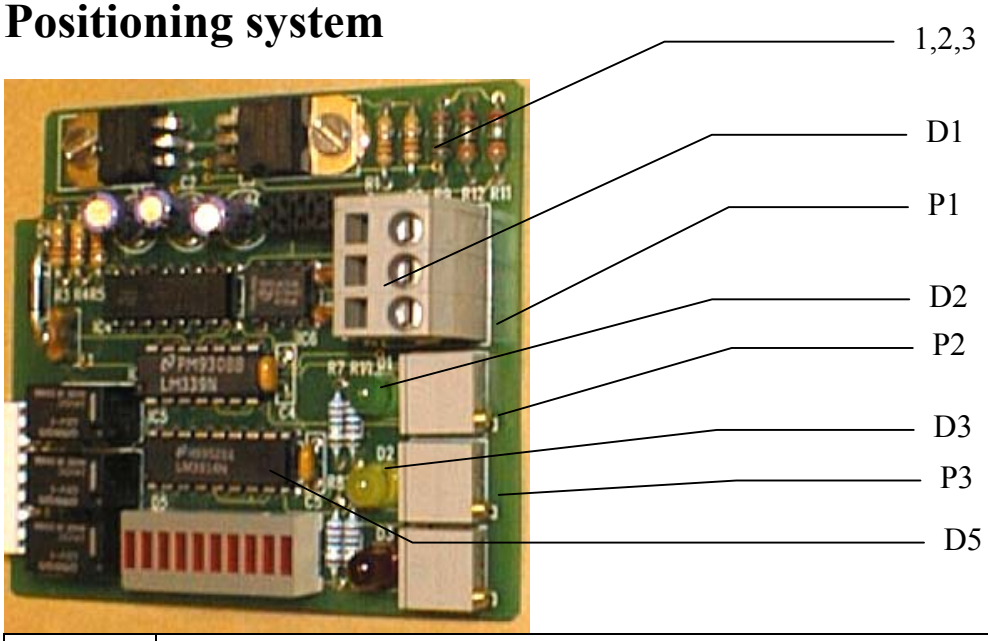
Other doors

Bring back the bottom beam and the door blade in the slot in the side frame.
If necessary reset, release mechanism in bottom beam. Then proceed with point 4 above.

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3. A-pos

Positioning system



NAME	DESCRIPTION
1	Input from sensor, wire 1.
2	Input from sensor, wire 2.
3	Input from sensor, wire 3.
P1	Door open position adjustment: By turning the P1 clockwise the top position for the door is set to a higher position. By turning the P1 counter clockwise the top position for the door is set to a lower position.
P2	Mid position adjustment: By turning P2 clockwise the MID position for the door is set to a higher position. By turning P2 counter-clockwise the MID position for the door is set to a lower position. If photocell in doorline is used, this position shall be set to 350 from floor level.
P3	Door closed position adjustment: By turning the P3 clockwise the closed position for the door is set to a higher position. By turning the P3 counter clockwise the top position for the door is set to a lower position.
D1, Green	Door open indication: This Led is lid when the door is below top position (set by P1).
D2, Yellow	Mid position indication: This Led is lid when the door is below 350 mm (set by P2).
D3, Red	Door closed indication: This Led is lid when the door is above closed position (set by P3).
D5	Door position: A moving point LED indicating the movement of the door.

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Adjustment of A-pos when out of limit

Instruction for adjustment of the A-pos when the potentiometer signal is outside the limit.

This occurs when the door doesn't reach the top limit position and the door stops and is not possible to close again without using the crank. All LED on the A-pos card becomes black. You are now on the last quarter of a turn on the potentiometer and a safety function has stopped the door.

- Take away the motor cover and the cover for the A-pos.
- Take away the A-pos potentiometer from the drive shaft.
- Bring the door to the open position.
- Turn the potentiometer to the end limit in the same direction as the door blade is rolled on to the top roll.
- Turn the potentiometer back one turn.
- Install the potentiometer on the drive shaft. Make it as straight as possible and so far in to the shaft that the cover can be installed without problems.
- Fix the cable without tensioning it – it must be enough cable for movement of the A-Pos potentiometer.
- Adjust down the top position
- Run the door and adjust the end positions.
- Install the cover for the A-pos.
- Install the motor cover.

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4. Maintenance

NOTE !

* Trained, qualified personnel should conduct all electrical work.

* Service in the door shall only be performed by personell with knowledge of the design of doors and the use of doors.

* Before maintenance work begins, mains current must be switched off at a lockable mains switch, and the switch marked with a sign indicating that work is in progress. Then check the power supply, to ensure that the electrical cabinet has no power.

* Never run the door while the frame doors are open.

* Before working on the door, make sure that all scaffolds, ladder etc. meet safety requirements, and are in good condition.

Inspection and maintenance that may be made by the customer

Door - all models

- Check safety functions as Emergency stop, safety edge, and photocell in doorline.
- Check that all warning signs are undamaged and well visible.
- Check that all shields and hoods are in correct place and haven't got any damages.
- Check that the controlbox is closed.

Safety edge

- Check the safety edge in an appropriate way, like hitting the safety edge, while the door is closing, with the hand.

Always contact the Albany Door Systems representative if there are any errors in the safety equipment.

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Other

Check the control equipment - for example buttons, photocells, radar etc. If necessary clean and adjust.

Lubricant

For steel cable: Brilube 30
For drive unit: Mobil Gear 627 or corresponding lubricant.



NOTE !

Spilled oil can cause accidents! Make sure the floor is properly cleaned.

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5. System overview

Albany Door Systems doors can be equipped with a number of different control devices, such as radar, radio, photocells, induction loops etc. The ACS 30 control system makes it possible to further customise the door, giving you the Albany Door Systems door that exactly meets your needs.

Following is a partial list of available features.

Control system

Standard	Prepared	Option	Feature
X			Time-delayed closing after impact on safety edge
X			Pause before reversing
X			Zero voltage protection
X			Supply voltage 3x230 or 3x400 or 3x415 VAC 50/60 Hz
		X	Flashing traffic lights
X			Manual reset after safety edge impact
X			Electrical safety edge
X			Pneumatic safety edge
X			Continuously supervised safety edge

Control equipment

Standard	Prepared	Option	Feature
X			Open/close buttons
X			Emergency stop button
		X	Open, Close button
		X	Pull switch
		X	External detector for opening loops
		X	External detector for loops to reverse the door
		X	Opening photocell
		X	Doorline photocell
		X	Extra doorline photocell
		X	External Radio
		X	Radar

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Appendix A

Address information

If you have any questions or need service, please contact your local Albany Door Systems representative

Notes

Date	Number of openings	Remarks

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Vehicle detector

IMPORTANT. Please read the Safety instructions in User's manual before you start the installation!

General

The loop is made of electrical wire 1.5 mm with a good insulation. Best result is achieved if the loop and the lead-in wire are not joined. If this can not be done it is very important to have a good and waterproof electrical joint.

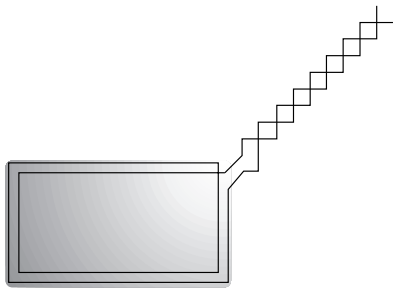
Loops with perimeters up to 12 meters should have 3 turns.

Loops with perimeters exceeding 12 meters should have 2 turns.

Installation

Wire between loop and control box

The lead-in wire is made of the same type of wire as the loop, and it should be twinned at least 10 turns per meter. You can also use a 2-part screen cable. The screen should be grounded in the detector end and not in the loop end.



Size of the loop

The loop size in the traffic direction should be at least twice the distance between the ground and the chassi of the highest vehicle which should traffic the opening, still at least 0.8 m length and 2.0 m width.
To be able to detect bicycles the loop should be 1.1 m long in traffic direction and be laid in a 45° direction against traffic direction.

Where to place the loop

To assure good function, please notice the following;

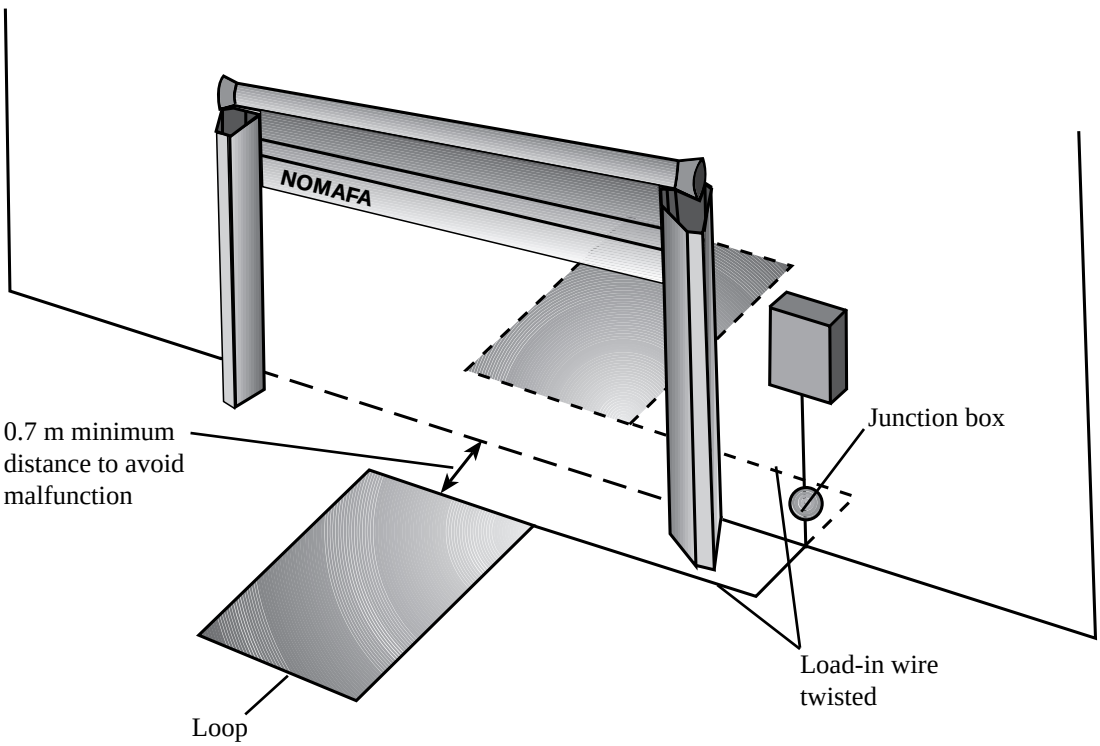
- Loops must be placed an adequate distance from moving doors and other metal objects, at least 0,7 m, to avoid malfunction.

- Minimum distance to reinforcement bars or other solid metal objects: 50 mm.

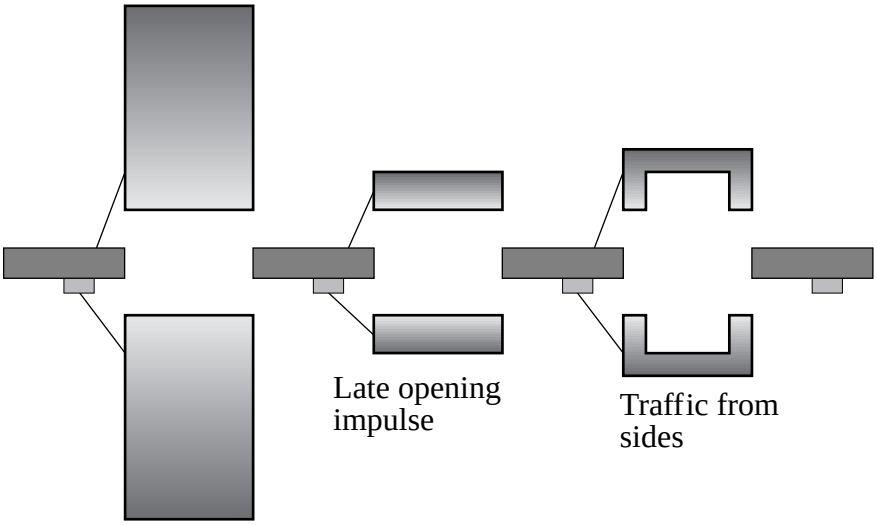
- Connections between the loop, lead-in wire and junctions have to be completely waterproof, also damages on insulation will cause an unstable magnetic field and unsure function.

- Heating cable, high voltage cables, and cables for AGV-trucks, are not allowed to be nearer the loop than 0.5 m.

- The loop track cannot contain other cables.



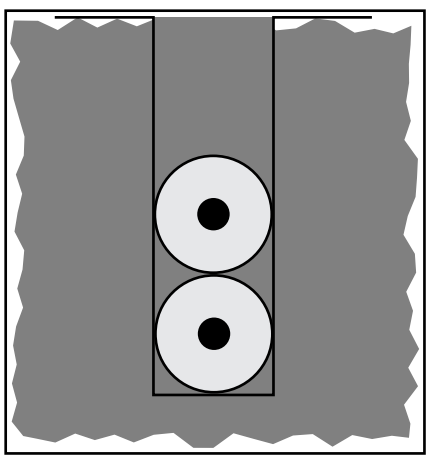
Examples on loop location for different requirements



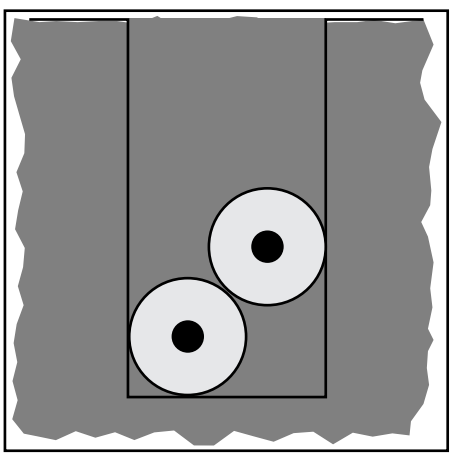
Installing the loop

Loops are normally wound into tracks cut in the road surface. These are typically (5-7 mm Wide) x (35-50 mm Deep) and back-filled with molten asphalt or concrete.

A loop track must not contain any other conductor than the loop itself.
Total feeder length is normally limited to 150 meters.
Loops which are to be installed in reinforced concrete should be as high as possible above the reinforcement bars, at least 50 mm, otherwise loss of sensitivity may occur.



CORRECT placement of loops in track.



WRONG!!

Test the loop

Extreme caution should be exercised when loops are to be installed near any type of heater. The use of loops and heaters together becomes more prohibitive the closer they are laid and impossible if the heating cables passes over or under the loop.

The resistance to earth at 500 V should be greater than 10 MΩ (normally 500 MΩ). Also measure the serial resistance which should be about 0.5 Ω, more than 5 Ω indicate a fault.

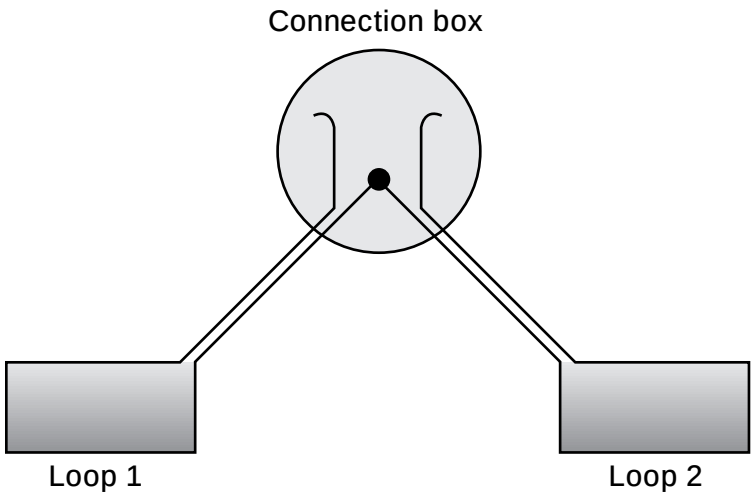
Before backfilling the track, test the system with actual vehicle!

Other

When a 2 loops configuration are operated by one detector, both loops shall be approximately the same size and shape.

The loops shall be connected in series.

When 2 detector systems are operating close to each other be sure to choose different frequency settings.



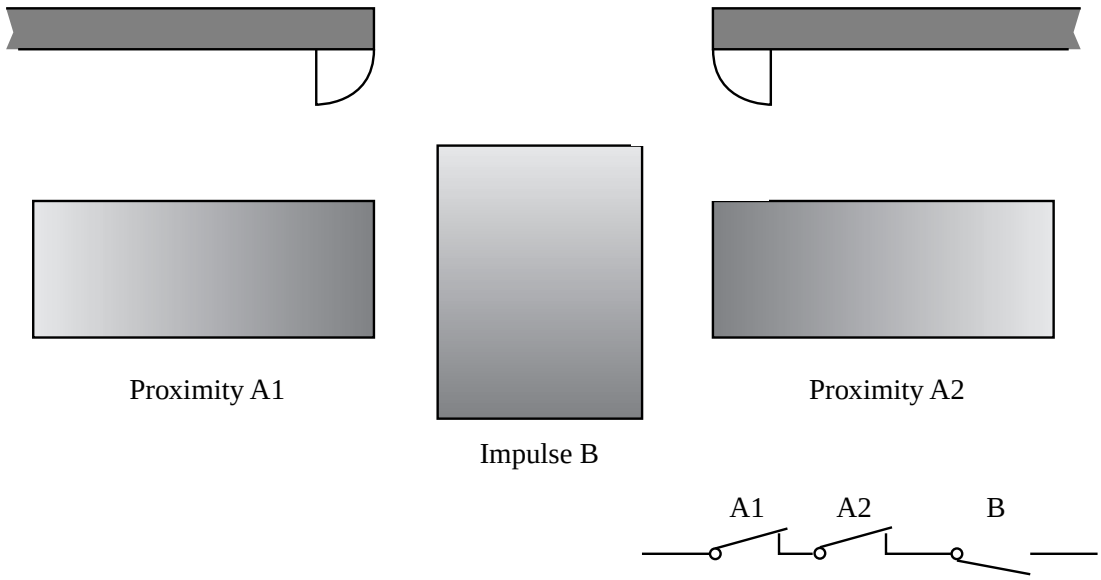
Special coupling loop

Cross-traffic interlock

This arrangement requires two detectors.

Function

The door will not open if loop A1 or A2 actuates before B.



Note!
Only impuls signals from loop B, A1 and A2 should be connected to the same detector.

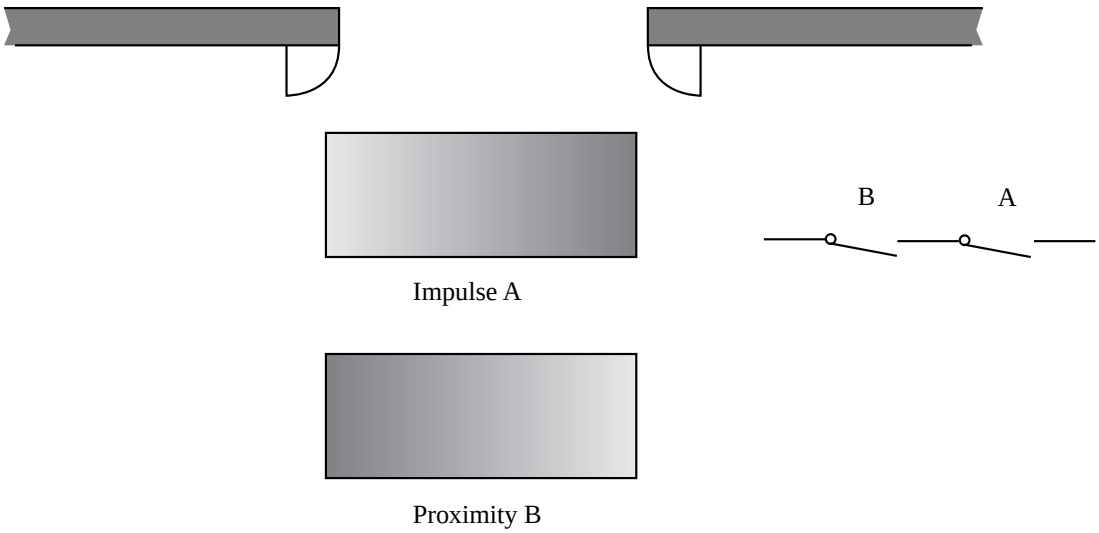
Even though a vehicle is on the loop, the door can begin to close. The special coupling loop should therfore be complemented by a photocell in doorline.

Direction sensitivity loops

This arrangement requires two detectors.

Function

In order to open the door loop B shall be actuated before loop A. Proximity signal loop B, impuls loop A.



Note!
Even though a vehicle is on the loop, the door can begin to close. The special coupling should therfore be complemented by a photocell in doorline.

Description of detector



Features

- Self-contained miniature boxed unit.
- 11 pin relay base connector.
- Presence and pulse output.
- Fast automatic tuning.
- Preset presence time.
- Choice of sensitivity level.
- Comprehensive fault detection.

The use of a microcomputer combines high performance with ease of use in a compact unit. Exceptional noise immunity and reliability enables this unit to function in normally unacceptable conditions.

All detection mode functions are set by means of a front panel rotary switch. Tuning is automatic and fast. Once tuned the detector will track all environmental drifts continuously. The selected presence time is constant and substantially independent of vehicle type. Timing is normally from first vehicle entry.

A loop inductance outside the tuning range of the detector is sensed as a fault. A fault will cause the LED to flash at a 1 Hz rate, with uneven mark-space ratio, and the presence relay to give a permanent presence call. The unit will attempt to retune until the fault is cleared. Whilst retuning the presence call is maintained.

Some vehicles (e.g. high chassis lorries with steel-braced wheels) produce an increase in inductance as they pass over the loop. This will cause some detectors to "lockup". The MZ range of detectors incorporates a feature to prevent this. Normal operation is unaffected.

How to read the loop detector scale

Frequency mode

Used when two detectors are installed close to each other and disturbance between them can appear. If one is switched on high and one on low scale they are not able to react on a wrong signal. If using only one detector it will not make any difference if it is set on high or low.

Presence time

Choose between 5 and 120 minutes. Function is used in case (as example) a forklift charging station is located close to the loop. In this case the door might be activated and stays open until the forklift is moved from the charging place. With this function the detector will accept the forklift as a dead material and close after 5 or 120 min. depending on the selected setting.

Sensitivity

Sensitivity can be made in four steps and depends on the type of forklift. The setting should be tested at place and with the vehicle that should pass.

Examples

Scale no. 3 - High frequence, 5 min, high sensitivity.

Scale no. 4 - High frequence, 120 min, low sensitivity.

Scale no. 12 - Low frequence, 120 min, low sensitivity.

FREQUENCY MODE		**HIGH**		**LOW**	
Presence Time (mins)		**5**	**120**	**5**	**120**
SENSITIVITY	Low	0	*(4)*	8	*(12)*
	Med. low	1	5	9	13
	Med. high	2	6	10	14
	High	*(3)*	7	11	15

Connections

Via 11 pin "socket" at rear of box.

PIN	Function
1	AC/DC +
2	AC/DC -
3	Pulse output, relay common
4	Pulse output, relay contact momentarily closed
5	Presence output, relay contact closed for detect
6	Presence output, relay common
7	Loop input
8	Loop input
9	Ground
10	Presence output, relay contact open for detect
11	Not used

Technical specifications

Tuning range
The tuning range depends on the frequency setting:
- 20 to 1000 µH (high frequency)

Tune time
Normally 2 seconds.

Operating temperature range
- 40° C to 80° C.

Reset
The detector is automatically reset when power is applied or the mode switch is adjusted. If required a reset can be affected by rotating the mode switch one (or more) positions and returning to the required setting.

Response time
Standard response time typically 100 - 150 ms (all sensitivities).

Supply
24 V AC/DC ± 20 % 120 mA max. Other voltages on request.

Relay Outputs
Presence: Changeover relay.

Pulse: Relay contact close for about 100 ms on detection.

Contact: 600 V, 5 A, 250 V AC, 30 V DC.

Dimensions
38 x 76 x 72 mm (W x H x D) excluding connector.

Weight
Approx. 135 g (24 V).

Lightning and transient protection
All inputs are transformer isolated, protected from over voltage on loop leads and flash over from internal circuit to ground.

Ordering information

MZ 24 Art. No. 96-7452-91

MD 1-1
USER'S GUIDE
MOTION SENSORS FOR INDUSTRIAL DOORS
RADAR : for high mounting

TECHNICAL SPECIFICATIONS

Technology : microwave and microprocessor
Transmitter frequency : 24.175 GHz
Transmitter radiated power : < 20 dBm EIRP
Transmitter power density : < 5 mW/cm²
Mounting height : from 3.5 to 7 m
Tilt angle : 0° to 180° in elevation
Detection zone (typical) : 4m (W) x 5m (D) for a mounting height of 5 m
Detection mode : movement
Minimum detection speed : 5 cm/s (measured in the sensor axis)
Supply voltage : 12V to 24V AC +/- 10%
12V to 24V DC +30% / -10%
Mains frequency : 50 à 60 Hz
Power consumption : < 2 W
Output relay : free of potential change-over contact
- Max contact voltage : 42V AC/ DC
- Max contact current : 1A (resistive)
- Max switching power : 30W (DC) / 60 VA (AC)

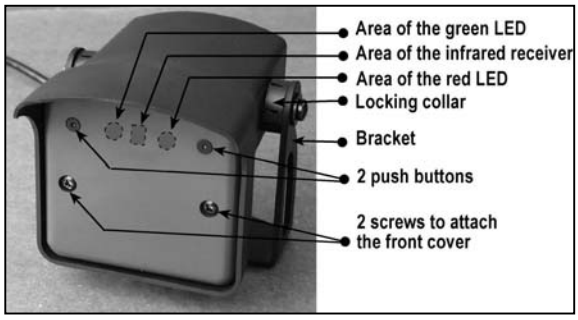
Hold time : 0.5s to 9s (adjustable)

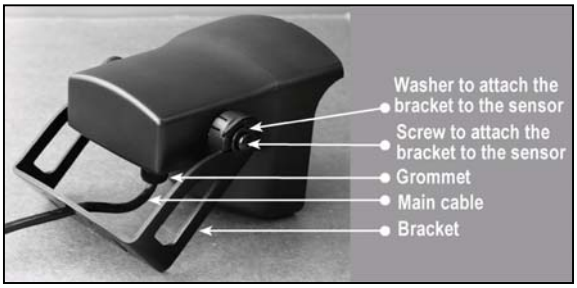
Manual adjustment :
- orientation of sensing field (mechanically)
- multiple functions (by push buttons).

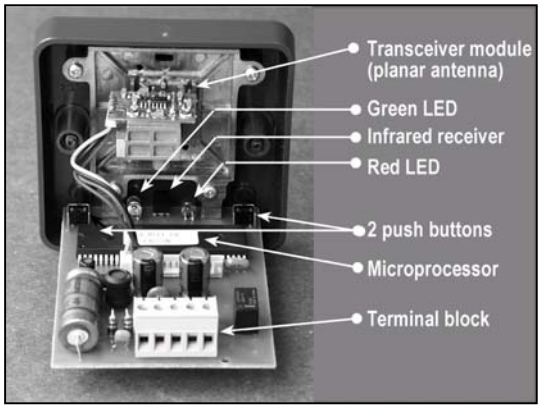
Remote control adjustments :
- Sensitivity.
- Hold time.
- Detection mode.
- Pedestrian and parallel traffic rejection mode.
- Relay configuration.

Temperature range : -30°C à +60°C
Degree of protection : IP65
Product conformity : R&TTE 1999/5/EC
EMC 89/336/EEC
Dimensions : 127mm (D) x 102 mm (W) x 96mm (H)
Weight : 400 g
Plastic of the casing : ABS and Polycarbonate
Bracket : black anodized aluminium
Cable length : 10 m
Cable diameter : 3 mm (minimum)
6.5 mm (maximum)

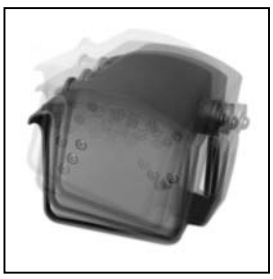
DESCRIPTION OF THE SENSOR







INSTALLATION TIPS



- The sensor must be firmly fastened in order not to vibrate.



- The sensor must not be placed directly behind a panel or any kind of material.

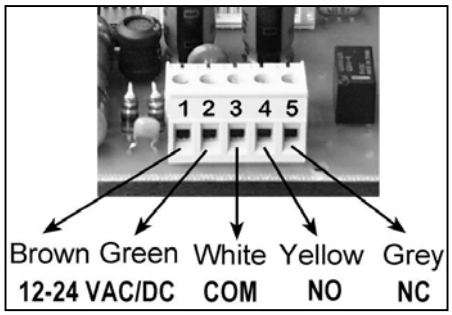


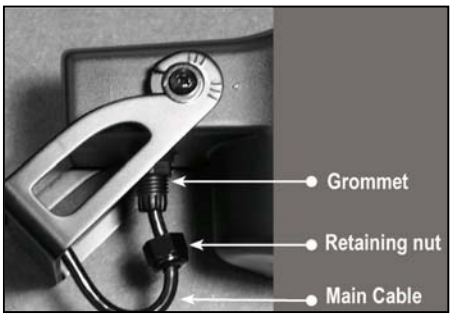
- The sensor must not have any object likely to move or vibrate in its sensing field.



- The sensor must not have any fluorescent lighting in its sensing field.

WIRING



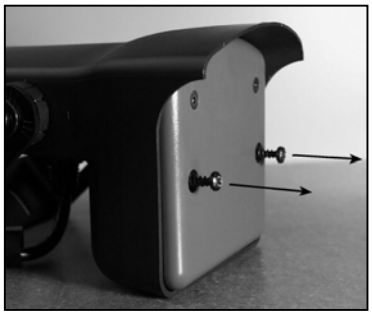


To remove or to insert the cable :

- Unscrew the retaining nut;
- Pass the cable through the grommet and the retaining nut.
- Tighten the retaining nut.

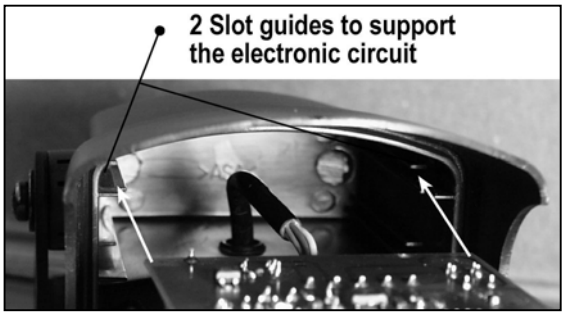
OPENING AND CLOSING THE SENSOR

Opening the sensor



- Loosen the retaining nut until the cable slides easily into the grommet ;
- Partially unscrew the 2 front cover screws ;
- Pull out the front cover with the 2 front cover screws.

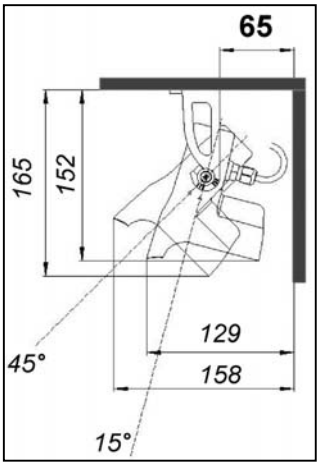
Closing the sensor



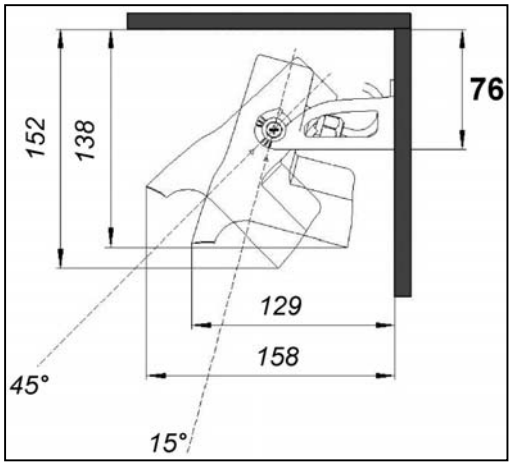
- Connect the quick disconnect terminal block to the main electronic circuit ;
- Slide the main electronic circuit into the 2 housing slot guides and pull the main cable gently at the same time ;
- Gently push the front cover and make sure that the external housing is properly seated (front cover must be flush with housing).
- Screw the 2 front cover screws and tighten the retaining nut.

SENSOR DIMENSIONS AND MOUNTING

Ceiling Mounting

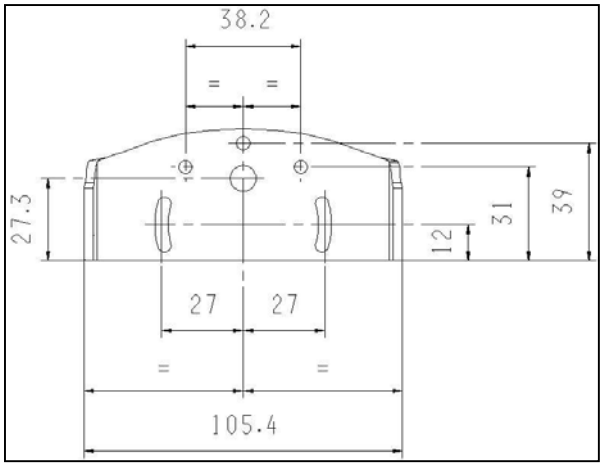


Wall Mounting

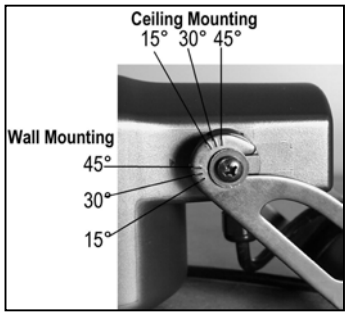


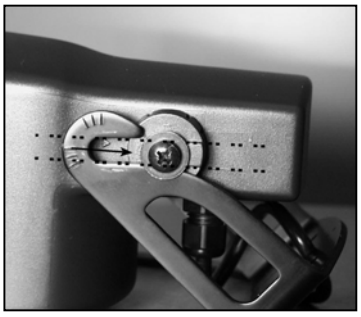
Remark : The bold-type values give the minimum distance required to be able to fully adjust the sensor.

Bracket dimensions



BRACKET MOUNTING



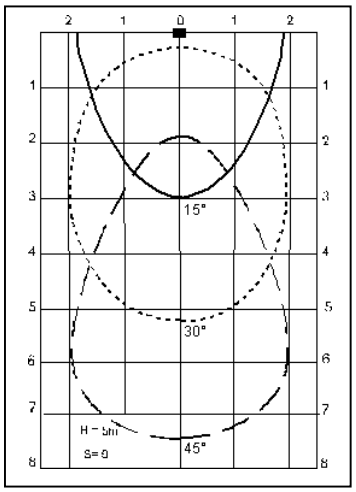


- Check that both locking collar are at the same angle;
- Align the bracket slot to the locking collar guide as shown.

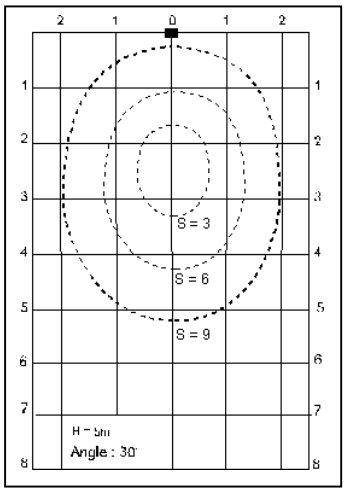
SETTING THE SENSING FIELD DIMENSIONS

MD 1-1 (Mounting height : 5m)

The sensing fields here on the right correspond to the following adjustments :
- tilt angle : 15°, 30°, 45° ;
- sensitivity : 9.

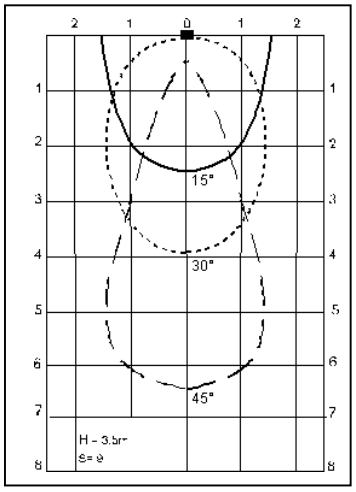


The sensing fields here on the right correspond to the following adjustments :
- tilt angle : 30°
- sensitivity : 9, 6, 3

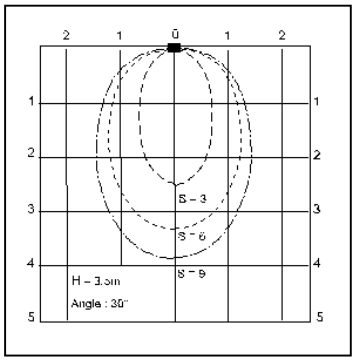


MD 1-1 (Mounting height : 3.5 m)

The sensing fields here on the right correspond to the following adjustments :
- tilt angle : 15°, 30°, 45° ;
- sensitivity : 9.



The sensing fields here on the right correspond to the following adjustments :
- tilt angle : 30°
- sensitivity : 9, 6, 3



LED SIGNAL

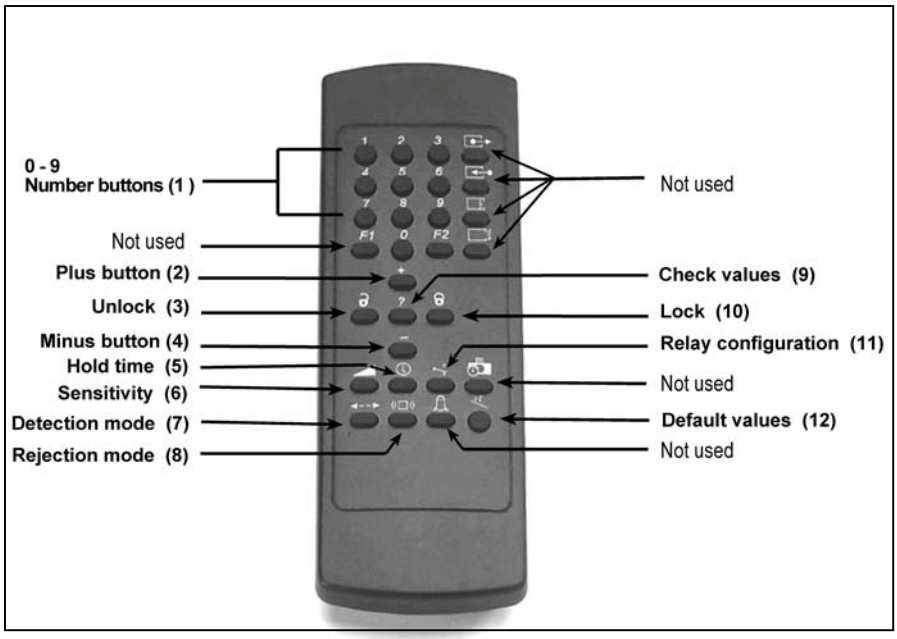
- When the power is turned ON, the red and green LEDs flash for few seconds.
- During a detection the red LED lights on.
- During configuration the red LED flashes a number of times corresponding to the parameter being changed (see next table). The green LED flashes a number corresponding to its setting.

FUNCTIONS CONFIGURATION WITH REMOTE CONTROL

1. DESCRIPTION OF THE INFRARED REMOTE CONTROL



- Open the battery compartment at the back of the remote control ;
- Insert two AAA batteries supplied with the remote control as shown beside ;
- Close the batteries compartment.

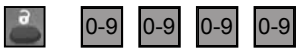


Remark : For optimum results point the remote control at the sensor before pressing its buttons.

2. CONFIGURATION OF THE SENSOR

Each setting change using the infrared remote control must start with an unlocking and end with a locking of the sensor. It is important to point out that any parameters changed using the remote control supersede any previous setting.

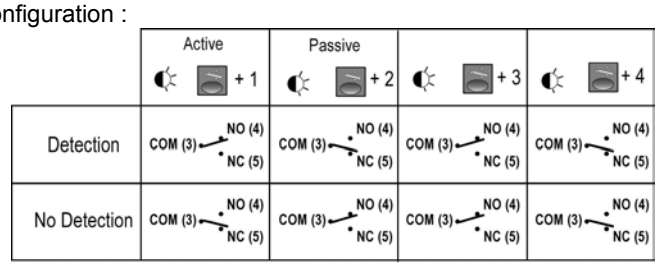
The table below lists all the parameters which can be adjusted with the remote control as well as the operations required to adjust them.

PARAMETER KEY	USER'S ACTIONS	FACTORY SETTING	LED SIGNAL
UNLOCK 	Press the **UNLOCK key (3).** Enter your four-digit access code using **NUMBER keys 0-9 (1)**. During the first sensor adjustment or if the access code is reset to the "0000" value (factory setting) or during the first minute after the power-on, press only the **UNLOCK key (3)** (no code required). *UNLOCK with code*  *UNLOCK without code* 	0000	The red LED flashes quickly waiting for the access code. After entering the correct code or if no code is required, the red LED flashes slowly to indicate that the unlock is successful and the adjustment session has begun. Note : = Adjustment session ON
LOCK 	When all the parameters have been set, press the **LOCK key (10)**. If you wish to enter a new access code, use **NUMBER keys 0-9 (1)** to enter the new four-figure code within 20 seconds. If no access code is entered or if you want to keep the current access code, press the **LOCK key (10)** once more. If no remote control key is pressed within 1 minute, the adjustment session is automatically locked. Pressing the LOCK key (10) twice within the first minute after the power-on reset automatically the access code to 0000 value. *LOCK with code change*  *LOCK without code or Code change* 	0000	The red LED stops flashing to return to its normal function.

Note : All parameters or functions listed in the following tables are only accessible if the sensor is in adjustment session. The RED LED is then slowly flashing.

During an adjustment session each parameter may be checked or changed at any time in the following way :	
PARAMETER KEY	**USER'S ACTIONS**
CHECK VALUES	Press the **key (5,6,7,8,11)** corresponding to the parameter to be checked and then press the **CHECK VALUES key (9)**. Count the number of times the green LED flashes, which corresponds to the value of the checked parameter. No green LED flash corresponds to the value 0. Repeat this operation to check the value of the other parameters if required. Example : SENSITIVITY key (6) – 7 flashes of the green LED : the sensitivity is set at the value 7. *CHECK VALUES* : X
PLUS	Press the **key (5 or 6)** corresponding to the hold time or sensitivity parameter to be modified. and then press the **PLUS key (2)** to increase the value by 1 unit. *PLUS* :
MINUS	Press the **key (5 or 6)** corresponding to the hold time or sensitivity parameter to be modified. and then press the **MINUS key (4)** to reduce the value by 1 unit. *MINUS* :

During a sensor adjustment session all the parameters may be reset to their factory values in the following way :	
PARAMETER KEY	**USER'S ACTIONS**
DEFAULT VALUES	Press the **DEFAULT VALUES key (12),** then press the **NUMBER key 1**. All the parameters are reset to the factory values (see below). *DEFAULTS VALUES :* 1

PARAMETER KEY	**USER'S ACTIONS**	**FACTORY SETTING**	**LED SIGNAL**
SENSITIVITY	Press the **SENSITIVITY key (6).** Use the **NUMBER keys 0-9 (1)** to enter the sensitivity required (or adjust this sensitivity using the **PLUS (2)** or **MINUS (4)** keys as explained above) *SENSITIVITY :* 0-9	7	The red LED flashes quickly waiting for the value. Once this has been entered, it flashes slowly again.
HOLD TIME	Press the **HOLD TIME key (5).** Use the **NUMBER keys 0-9 (1)** to enter the required hold time (0.5 s to 9 s) (or adjust this parameter using the **PLUS (2)** or **MINUS (4)** keys as explained above). *HOLD TIME* : 0-9	0.5 s	The red LED flashes quickly waiting for the value. Once this has been entered, it flashes slowly again.
RELAY CONFIGURATION	Press the **RELAY CONFIGURATION key (11).** Use the **NUMBER keys 1-4 (1)** to select the required relay configuration : 	1 (Active Output)	The red LED flashes quickly waiting for the value. Once this has been entered, it flashes slowly again.
DETECTION MODE	Press the **DETECTION MODE key (7).** Use the **NUMBER keys 1-3 (1)** to select the required mode : key 1 : bidirectional key 2 : unidirectional approach key 3 : unidirectional depart *DETECTION MODE :*	2 (Unidirectional Forward)	The red LED flashes quickly waiting for the value. Once this has been entered, it flashes slowly again.

<table>
<tr>
<td>REJECTION MODE
  </td>
<td>Press the REJECTION MODE key (8).
Perturbations immunity is used to avoid detection due to environmental perturbations (vibrations, rains, etc).
“Pedestrian/parallel traffic rejection” provides both rejection of pedestrian and rejection of any parallel traffic simultaneously.

Use the NUMBER keys 1- 5 (1) to enter the required rejection mode :
key 1 : detection of all kind of targets in motion
key 2 : detection of all kind of targets in motion
+ perturbations immunity
key 3 : Low “Pedestrian/parallel traffic” rejection
+ perturbations immunity
key 4 : Mid “Pedestrian/parallel traffic” rejection
+ perturbations immunity
key 5 : High “Pedestrian/parallel traffic” rejection
+ perturbations immunity

The discrimination between a pedestrian and the different vehicles depends mainly on the mounting height and the microwave module tilt angle. Be careful that the rejection function increases the response time of the sensor.

Use the next table as suggestions and do not hesitate to increase or decrease the rejection level to obtain the required rejection.
<table>
<tr><td colspan="2" rowspan="2">Recommended key</td><td colspan="3">Tilt angle</td></tr>
<tr><td>15°</td><td>30°</td><td>45°</td></tr>
<tr><td rowspan="3">Mounting height</td><td>7 m</td><td>3</td><td>4</td><td>4</td></tr>
<tr><td>5 m</td><td>3</td><td>4</td><td>4</td></tr>
<tr><td>3.5 m</td><td>4</td><td>4</td><td>5</td></tr>
</table>
</td>
<td>1
(No rejection)</td>
<td>The red LED flashes quickly waiting for the value.
Once this has been entered, it flashes slowly again.</td>
</tr>
</table>

FUNCTIONS CONFIGURATION WITH PUSH BUTTONS

Without remote control all of the parameters can be set using the 2 push buttons.
Note : These two buttons are accessible from the sensor front cover with a small point.

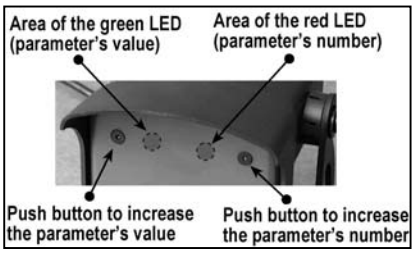
- To adjust the sensor using the buttons :
  Press and hold either push button for 2 seconds (until the LEDs flash) and then release the button ;
- To end adjusting the sensor using the buttons :
  Press and hold either push button for 2 seconds (until the LEDs stop flashing) and then release the button;
  Note : if no button is pressed within 20 seconds the adjustment session is automatically ended.
- To reset all the parameters to the factory values :
  Press and hold both push buttons simultaneously until the two LEDs switch on for 1 second (after 2 seconds).

During the manual adjustment session the red and green LEDs flash successively and continuously.
- The flashing number of the **red LED** provides the **number of the displayed parameter** (see next table);
- The flashing number of the **green LED** provides the **value of the displayed parameter** ;

Note : no red flashing indicates the zero value.

During the manual adjustment session :
- Each press on the **right button increases the number of the displayed parameter** by one unit ;
- Each press on the **left button increases the value of the displayed parameter** by one unit.

Note : When the maximum value or the highest number of the parameter is reached, it will return to its minimum values.



Parameter Number	Parameter	Values	Factory setting
1	Sensitivity	0 - 9	7
2	Hold time	0 - 9	0
3	Relay configuration	1 - 4	1
4	Detection mode	1 - 3	2
5	Pedestrian rejection mode	1 - 5	1

For example, to change the sensitivity from 7 to 9 and the rejection mode from “detection of all kind of targets in motion” to High “Pedestrian/parallel traffic” rejection.

- Press any button for 2 seconds to enter the adjustment session and then release it.
- The red LED flashes once (parameter 1 = sensitivity) and the green LED flashes 7 times (sensitivity=7).
- Press the left button twice to increase the sensitivity from 7 to 9.
- The red LED still flashes once (parameter 1 = sensitivity) but the green LED flashes 9 times now (sensitivity=9).
- Now press the right button 4 times to move to function 5 (rejection mode) ;
- The red LED flashes 5 times (parameter 5 = rejection mode) and the green LED flashes once (“detection of all kind of targets in motion"
- Press the left button 4 times to set the parameter to High “Pedestrian/parallel traffic” rejection.
- The red LED still flashes 5 times (parameter 5 = rejection mode) but the green LED flashes 5 times now (High “Pedestrian/parallel traffic” rejection).
- Press any button during 2 seconds to end the adjustment session and then release it.

TROUBLE-SHOOTING

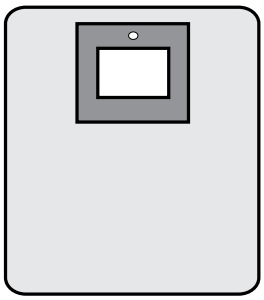
SYMPTOM	***PROBABLE CAUSE***	***CORRECTIVE ACTION***
The door will not open and the red LED does not light up.	The sensor power is off.	Check power supply. Check the supply voltage.
The door opens and closes constantly.	The sensor "sees" the door moving.	Increase the tilt angle and/or reduce the sensitivity.
	When closing the door creates vibrations picked up by the sensor.	Ensure that the sensor is correctly fixed. If the rejection mode is set at level 1, set this parameter to level 2. Reduce the sensitivity. Switch to unidirectional mode.
The door opens and closes after a given time for no apparent reason.	The sensor is picking up unintended traffic motion.	Reduce the sensitivity. Reduce the tilt angle.
The sensor is not capable of activation near the door.	The tilt angle is too large.	Reduce the tilt angle.
The sensor does not respond to the remote control.	The batteries are weak.	Check the batteries insertion. Change the batteries.
	The access code has been changed.	• Press both push buttons simultaneously to reset all the parameters to the factory values. • Or switch off the power supply. Within the first minute after the power on, change the access code.

IMPORTANT. Please read the Safety instructions in User's manual before you start the installation!

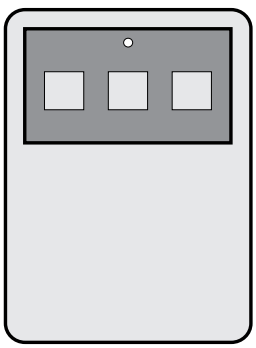
General

The system transmitts at a frequency of 40 MHz in the AM band. Telecoms reserves these frequencies for this type of short-distance transmissions. A radio signal is sent from the hand- or the vehicle- transmitter to a receiver located close to the rapid roll door. A relay in the reciever activates and the door opens.

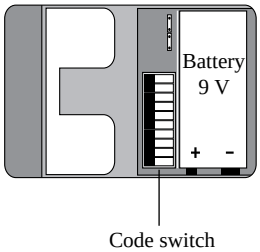
Hand transmitter



One push button for opening of one door.
Type 201

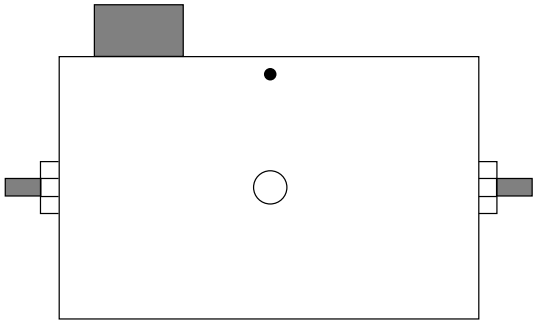


Three push buttons for opening of three different doors or one door with UP - STOP - DOWN.
Type 203

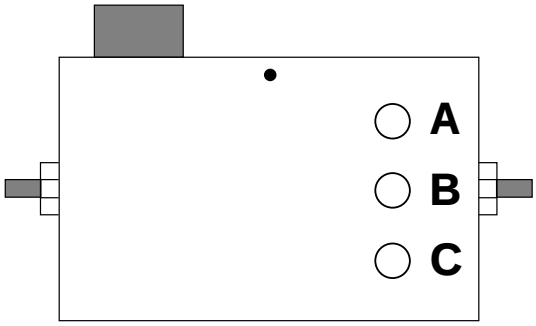


Transmitter is supplied from a 9 V battery placed under a cover on the rear side of the hand transmitter. Life time of battery at normal use is between 3 - 12 month.

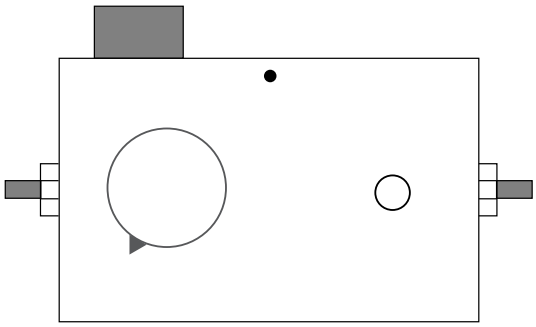
Vehicle transmitter



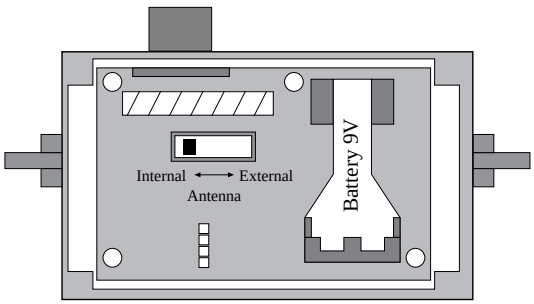
One push button for opening of one door with impulse function.
Type S300 1B



Three push buttons for opening of three different doors or one door with
UP - STOP - DOWN.
Type S300 3B

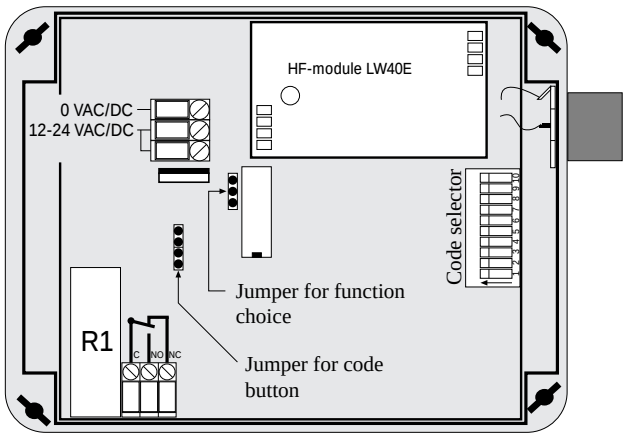


One push button and a rotary switch from 1 - 10 permitting control of up to 10 doors.
Type S300 3B

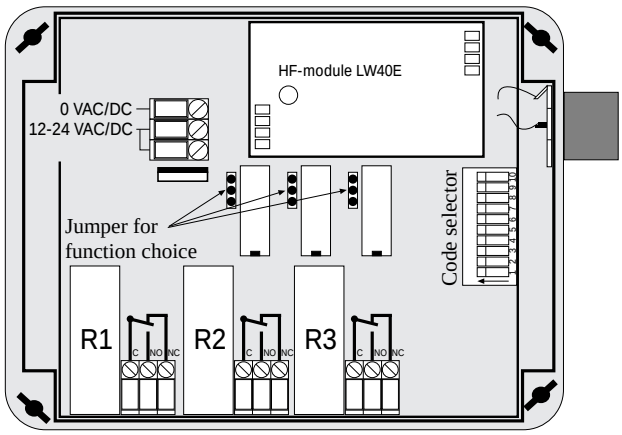


Transmitter is supplied from a 9 V battery placed under the front cover. Life time of battery at normal use is between 3 - 12 month. This transmitter is also available for external 10 - 35 V DC power supply.

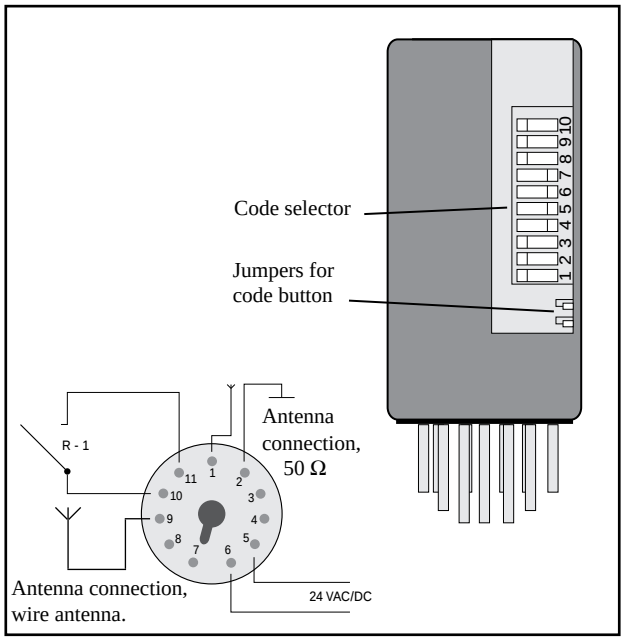
Receivers



One channel external receiver
Type 7344
(24 V AC/DC)
One relay output.



Three channel external receiver
Type 7346
(24 V AC/DC)
Three relay output.



One channel internal receiver
Type 7314, (24 V AC/DC). All connections are made on a standard 11-pole socket. One relay output.

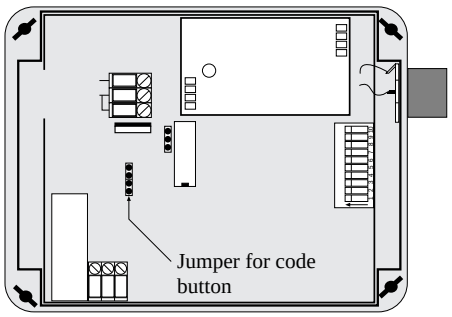
The antenna can be connected in two different ways:
1. Coaxial cable connected to No. 1 and the screen to No. 2.
2. Wire antenna or coaxial cable connected to No. 9 (the screen should not be connected).
Note! With the antenna connected according to item 2, it´s possible to have longer operating range but the receiver will be more sensitive for disturbances.

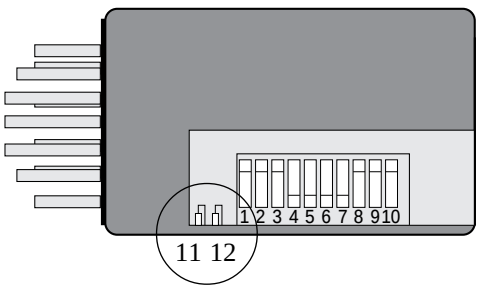
Coding the transmitter and receiver

The radio system must be coded at the time of installation, so that the system's transmitters and receivers all function logically. Consider these examples. The first applies for multi-channel receivers, the second for single-channel receivers.

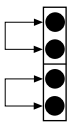
Multi-channel receivers
Three receivers, numbered ONE, TWO and THREE, mounted in three separate locations. With a 10-channel transmitter, the respective receivers should be coded so that receiver ONE functions when the transmitter's dial is set in one-position, and so on.

Single-channel receivers
Four single-channel receivers, and a three-button transmitter. In this case, transmitter and receiver code switches are set identically. Only the respective receivers' jumpes are adjusted. As a result, each receiver will function for a different button on the transmitter. Illustrated below are the settings for the receiver's codes for respective transmitter buttons.

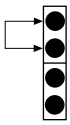




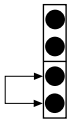
Jumper position for selection of transmitter button to the 1st receiver.



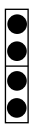
Jumper position for selection of transmitter button to the 2nd receiver.



Jumper position for selection of transmitter button to the 3rd receiver.



Jumper position for selection of transmitter button to the 4th receiver.



Connect both jumpers 11 and 12 to receive signals from the 1st button on the transmitter.

Connect jumper 11 to receive signals from the 2nd button on the transmitter.

Connect jumper 12 to receive signals from the 3rd button on the transmitter.

Disconnect both jumpers 11 and 12 to receive signals from the 4th button on the transmitter.

Code key

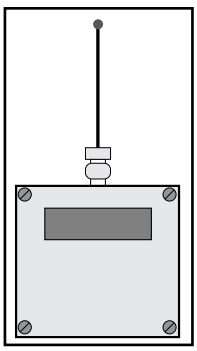
10-channel rotary switch vehicle transmitter and hand transmitter

Decide which receiver should function when the transmitter is set to position '1', position '2', and so on.
Each transmitter contains a 6-pole code switch which determines the user's security code. Set one or more of the switches to the ON position at the transmitter's code selector. These settings must be identical on all the system's transmitters.
Remove the cover from each receiver, exposing the 10-pole code selector on the printed circuit card. Set switches 1-6 in the same positions as on the transmitters. (Example: transmitter switches 1 and 5 are set to ON. Therefore, set receiver switches 1 and 5 to ON, and 2, 3, 4 and 6 to OFF.)
Security codes are now set. Now we shall set receiver codes to function with the transmitter's different channels. Using this code key, up to ten different 3-channel receivers can be set to function with one or more multi-channel transmitters.
Use a two-digit combination to number the different receivers from 01 to 10, corresponding to different transmitter channels. Then set respective receivers' code switches to match the transmitters' channel selector positions, as illustrated below.

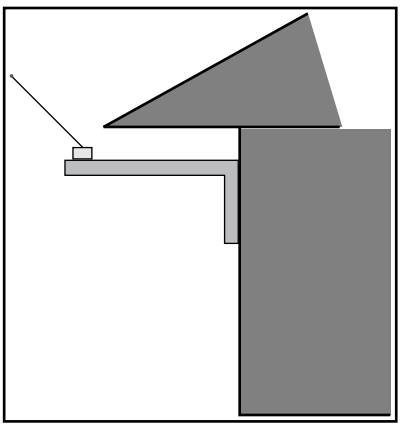
Rotary switch position: **(transmitter channel)**	**Switch set to ON position**	**Switch set to OFF position**
1	10	7, 8, 9
2	9	7, 8, 10
3	9, 10	7, 8
4	8	7, 9, 10
5	8, 10	7, 9
6	8, 9	7, 10
7	8, 9, 10	7
8	7	8, 9, 10
9	7, 10	8, 9
10	none	7, 8, 9, 10

Antennas and their position

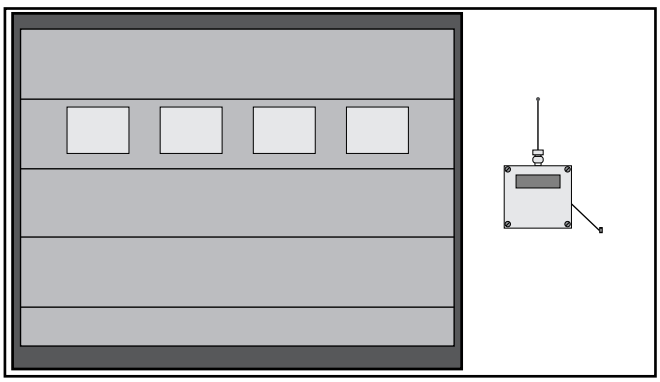
In a radio control system, all receivers should be equipped with some kind of antenna. It is the receiver's antenna which, principally, decides the radio installation's range.
The vehicle's transmitter should also have an antenna if it's placed in an area with a lot of obstructions. Inside the transmitter there is a switch for internal or external antenna. As a rule, one could say that the transmitter and the receiver should "see" each other. If they can't, one of them has to be relocated. The antenna should have an angle of 45 °. It is important that the antenna has a good earth connection.
Here are a few different types of antenna positions:

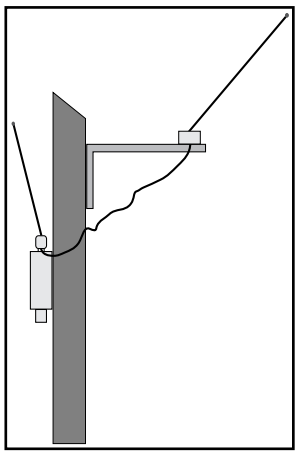


Antenna mounted directly on the wall receiver or directly on the control box if the receiver is built in.



Antenna with cable mounted on a wall with a bracket.





Wall receiver equipped with an inner and an outer antenna.

Technical specifications

Transmitter	
Frequency	40.685 MHz
Amplified modulated pulses	
Power supply hand type	9 V
Power supply mobile type	9 V or external 10 - 35 V DC
Power consumption	45 mA
Receiver	
Sensitivity	1.5 µV at 10 dB
Power supply	24 V AC/DC. Other voltages on request.
Power consumption	25 mA
Relay output	250 VAC, 10 A

Ordering information

Hand transmitters	**Type 201,** Art. No. 96-7360-85 **Type 203,** Art. No. 96-7362-85
Vehicle transmitters	**Type S300 1B,** Art. No. 96-7362-85 **Type S300 3B,** Art. No. 96-7368-85 **Type S300 10B,** Art. No. 96-7369-85
Receivers	**Type 7344,** Art. No. 96-7373-85 **Type 7346,** Art. No. 96-7377-85 **Type 7314,** Art. No. 96-7371-89